Quadratic residues

L5-NT1 Level 5 Advanced · Number theory

Kevin Hu
November 2026

Legendre symbols, Euler's criterion, the laws for -1 and 2, quadratic reciprocity, primes dividing quadratic forms and sums of two squares.
By the end you can you will be able to evaluate Legendre symbols with Euler's criterion and quadratic reciprocity.; you will be able to determine which primes divide values of a quadratic, and use this to rule out solutions of equations.; you will be able to use the two squares theorem and character sums in olympiad proofs..
Assumes: Orders, Fermat and Euler (L4-NT1), Olympiad Diophantine equations (L4-NT2). **Contests:** BMO2, EGMO, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

Let n be odd and let p be a prime factor of 2^n+1. Which residues can p have modulo 8? Orders (L4-NT1) only tell us that $\operatorname{ord}_p(2)$ divides $2n$ but not n, and that says nothing about p modulo 8. Instead, multiply $2^n \equiv -1$ by 2. Since $n+1$ is even,

$$-2 \equiv 2^{n+1} = \left(2^{(n+1)/2}\right)^2 \pmod{p},$$

so -2 is a square modulo p. The question is now: for which primes is -2 a square? This unit answers that question with any integer in place of -2. Here the answer (Theorem 4 below) is $p \equiv 1$ or $3 \pmod 8$. For example $2^5+1 = 3\cdot 11$ and $2^7+1 = 3\cdot 43$, and 3, 11, 43 are all 3 modulo 8.

Throughout, p denotes an odd prime.

1.1 Residues and the Legendre symbol

An integer a with $p \nmid a$ is a **quadratic residue** modulo p if $x^2 \equiv a \pmod p$ has a solution, and a **non-residue** otherwise. The **Legendre symbol** $\left(\frac{a}{p}\right)$ is 1 for a residue, -1 for a non-residue and 0 if $p \mid a$. With this convention the congruence $x^2 \equiv a$ has exactly $1+\left(\frac{a}{p}\right)$ solutions modulo p for every a, since a solution $x \not\equiv 0$ comes with its partner $-x \not\equiv x$.

Theorem 1.
Exactly $\frac{p-1}{2}$ of the numbers $1, 2, \ldots, p-1$ are quadratic residues, namely the residues of $1^2, 2^2, \ldots, \left(\frac{p-1}{2}\right)^2$.

Proof. $x^2 \equiv y^2$ if and only if $p \mid (x-y)(x+y)$, that is, $x \equiv \pm y$. So x and $p-x$ give the same square and the listed squares are distinct. □

Kevinvariant
kevinvariant08.github.io

Theorem 2 (Euler's criterion).
If $p \nmid a$ then $a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod p$. Consequently $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$ for all a, b.

Proof. Let $t = a^{(p-1)/2}$. By Fermat $t^2 \equiv 1$, so $t \equiv \pm 1$. If $a \equiv b^2$ then $t \equiv b^{p-1} \equiv 1$. Hence the $\frac{p-1}{2}$ residues are all roots of $x^{(p-1)/2} - 1$, which by Lagrange's theorem (L4-NT1) has at most $\frac{p-1}{2}$ roots modulo p. So no non-residue is a root, and $t \equiv -1$ for a non-residue. For the second statement, if $p \mid ab$ both sides are 0; otherwise both sides are congruent to $(ab)^{(p-1)/2}$ and lie in $\{-1, 1\}$, and $p > 2$. □

The multiplicativity is the part I use most: it reduces every symbol to $\left(\frac{-1}{p}\right)$, $\left(\frac{2}{p}\right)$ and $\left(\frac{q}{p}\right)$ for odd primes q. In particular the product of two non-residues is a residue.

1.2 The residues -1 and 2

Theorem 3.
$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$, so -1 is a residue if and only if $p \equiv 1 \pmod 4$.
Theorem 4. $\left(\frac{2}{p}\right) = 1$ if and only if $p \equiv \pm 1 \pmod 8$. Hence $\left(\frac{-2}{p}\right) = 1$ if and only if $p \equiv 1$ or $3 \pmod 8$.

Theorem 3 is Euler's criterion with $a = -1$. Note that it reproves the fact from L4-NT1 that odd prime factors of $n^2 + 1$ are 1 modulo 4.

Proof of Theorem 4 (compare two expressions for one product). Let $h = \frac{p-1}{2}$ and multiply the even numbers $2, 4, \ldots, 2h$: the product is $2^h\, h!$. Now replace each even number $2j > h$ by $2j - p$, which is negative with odd absolute value $p - 2j \le h$. The even numbers $2j \le h$ are all the even numbers in $[1, h]$, and the values $p - 2j$ are distinct odd numbers in $[1, h]$; there are $h - \lfloor h/2 \rfloor$ of them, which is the number of odd numbers in $[1, h]$. So the absolute values are $1, 2, \ldots, h$, each once, and

$$2^h\, h! \equiv (-1)^m\, h! \pmod p, \qquad m = h - \lfloor h/2 \rfloor.$$

Cancelling $h!$ and using Euler's criterion, $\left(\frac{2}{p}\right) = (-1)^m$. For $p = 8k+1, 8k+3, 8k+5, 8k+7$ we get $m = 2k$, $2k+1$, $2k+1$, $2k+2$. The statement about -2 follows from multiplicativity and Theorem 3. □

This settles the opening problem. Another application: for $k \ge 2$, every prime factor p of $2^{2^k} + 1$ satisfies $p \equiv 1 \pmod{2^{k+2}}$. Indeed L4-NT1 gives $\operatorname{ord}_p(2) = 2^{k+1}$, so $p \equiv 1 \pmod 8$, so 2 is a residue and $2^{(p-1)/2} \equiv 1$; hence $2^{k+1} \mid \frac{p-1}{2}$. (Euler's factor 641 of $2^{32} + 1$ is $5 \cdot 2^7 + 1$.)

Example 1.
Let p be an odd prime such that $q = 4p + 1$ is also prime. Prove that q divides $2^{2p} + 1$, and that 2 has order $q - 1$ modulo q.

Solution.
Note that $2p = \frac{q-1}{2}$, so by Euler's criterion $2^{2p} \equiv \left(\frac{2}{q}\right) \pmod q$. The first claim is therefore the statement that 2 is a non-residue modulo q, and Theorem 4 turns this into a question about q modulo 8. Since p is odd, $4p \equiv 4 \pmod 8$, so $q \equiv 5 \pmod 8$ and $\left(\frac{2}{q}\right) = -1$. Hence $2^{2p} \equiv -1 \pmod q$.
Let $d = \operatorname{ord}_q(2)$. Then $d \mid 4p$ and $d \nmid 2p$, so $d = 4$ or $d = 4p$. If $d = 4$ then $q \mid 2^4 - 1 = 15$,

Kevinvariant
kevinvariant08.github.io

impossible since $q \geq 13$. Hence $d = 4p = q - 1$. For example $p = 3$ gives $2^6 + 1 = 65 = 5 \cdot 13$.

1.3 Quadratic reciprocity

Theorem 5 (quadratic reciprocity).
For distinct odd primes p and q,

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

That is, $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$ unless $p \equiv q \equiv 3 \pmod 4$, in which case $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$.

I will not prove this here. Gauss published several proofs; the standard one at our level extends the argument of Theorem 4 (this is Gauss's lemma) and then counts lattice points in a rectangle. See Ireland and Rosen, *A Classical Introduction to Modern Number Theory*, Chapter 5, or Niven, Zuckerman and Montgomery, *An Introduction to the Theory of Numbers*, Section 3.2. In an olympiad you may quote Theorems 2 to 5 without proof.

To evaluate $\left(\frac{a}{p}\right)$: factor a, split by multiplicativity, flip each odd prime with reciprocity, reduce, repeat. For example, to see that 5 is a square modulo the prime 2029, note that $\left(\frac{5}{2029}\right) = \left(\frac{2029}{5}\right) = \left(\frac{4}{5}\right) = 1$. Three general formulas, for odd primes p not dividing the top, are worth remembering:

- $\left(\frac{3}{p}\right) = (-1)^{(p-1)/2}\left(\frac{p}{3}\right)$, which is 1 if and only if $p \equiv \pm 1 \pmod{12}$;
- $\left(\frac{-3}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{3}{p}\right) = \left(\frac{p}{3}\right)$, which is 1 if and only if $p \equiv 1 \pmod 3$;
- $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right)$, which is 1 if and only if $p \equiv \pm 1 \pmod 5$.

For the first, if $p \equiv 1 \pmod 4$ we need $\left(\frac{p}{3}\right) = 1$, so $p \equiv 1 \pmod 3$; if $p \equiv 3 \pmod 4$ we need $\left(\frac{p}{3}\right) = -1$, so $p \equiv 2 \pmod 3$. Combining gives $p \equiv 1$ or $11 \pmod{12}$.

Residues enter problems mostly through the following observation.

Theorem 6.
Let $f(n) = an^2 + bn + c$ have integer coefficients, let $D = b^2 - 4ac$, and let p be an odd prime with $p \nmid a$. Then p divides $f(n)$ for some integer n if and only if $\left(\frac{D}{p}\right) \neq -1$.

Proof. $4af(n) = (2an + b)^2 - D$, and $2an + b$ runs through all residues modulo p as n does, because $2a$ is invertible. □

Example 2.
Find all primes p for which there is an integer n with $p \mid n^2 + n + 2$.

Solution.
Theorem 6 applies to odd p, so deal with $p = 2$ first: $n^2 + n = n(n+1)$ is even, so $n^2 + n + 2$ is always even and $p = 2$ works. For odd p, $D = 1 - 8 = -7$, so we need $\left(\frac{-7}{p}\right) \neq -1$. The case $p = 7$ works ($n = 3$ gives 14).

Kevinvariant
kevinvariant08.github.io

For $p \neq 2,7$, reciprocity gives $\left(\frac{7}{p}\right) = (-1)^{\frac{p-1}{2}\cdot 3}\left(\frac{p}{7}\right) = (-1)^{(p-1)/2}\left(\frac{p}{7}\right)$, and $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$. The two signs cancel: $\left(\frac{-7}{p}\right) = \left(\frac{p}{7}\right)$. The squares modulo 7 are $1, 2, 4$.
So the answer is $p = 7$ together with all primes $p \equiv 1, 2, 4 \pmod 7$. This list already contains $p = 2$, since $2 \equiv 2 \pmod 7$.

1.4 Sums of two squares

Theorem 7 (Fermat).
Every prime $p \equiv 1 \pmod 4$ is a sum of two squares.
Lemma 8. If $q \equiv 3 \pmod 4$ is prime and $q \mid x^2 + y^2$, then $q \mid x$ and $q \mid y$.
Theorem 9. A positive integer n is a sum of two squares if and only if every prime $q \equiv 3 \pmod 4$ divides n to an even power.

Proof of Theorem 7 (Thue's lemma, a pigeonhole argument). By Theorem 3 there is r with $r^2 \equiv -1 \pmod p$. Let $m = \lfloor \sqrt{p} \rfloor$; then $m < \sqrt{p}$ since p is not a square. There are $(m+1)^2 > p$ pairs (a, b) with $0 \leq a, b \leq m$, so two different pairs give the same residue of $a + br$, say $a + br \equiv a' + b'r$. Put $u = a - a'$ and $v = b' - b$. Then $u \equiv rv$, so $u^2 \equiv r^2v^2 \equiv -v^2$ and $p \mid u^2 + v^2$. Also $(u, v) \neq (0, 0)$ and $|u|, |v| \leq m < \sqrt{p}$, so $0 < u^2 + v^2 < 2p$. Hence $u^2 + v^2 = p$. □

Proof of Lemma 8. If $q \nmid y$ then $(xy^{-1})^2 \equiv -1 \pmod q$, contradicting Theorem 3. So $q \mid y$, and then $q \mid x^2$, so $q \mid x$. □

Proof of Theorem 9. If the condition holds, n is a product of factors $2 = 1^2 + 1^2$, primes $p \equiv 1 \pmod 4$ (Theorem 7) and squares $q^2 = q^2 + 0^2$, and the identity $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$ shows that products of sums of two squares are sums of two squares. Conversely suppose $n = x^2 + y^2$ and $q \equiv 3 \pmod 4$ divides n. By Lemma 8, $q \mid x, y$, so $n/q^2 = (x/q)^2 + (y/q)^2$ is again a sum of two squares, with v_q smaller by 2. Repeating, we reach a sum of two squares with v_q equal to 0 or 1; the value 1 is impossible by the same argument, so $v_q(n)$ is even. □

Example 3.
Prove that the equation $y^2 = x^3 + 23$ has no integer solutions.

Solution.
Small moduli first. If x is even then $y^2 \equiv 23 \equiv 7 \pmod 8$, impossible. If $x \equiv 3 \pmod 4$ then $x^3 \equiv 3$ and $y^2 \equiv 26 \equiv 2 \pmod 4$, impossible. But $x \equiv 1 \pmod 4$ is not ruled out modulo 8 ($x = 1$ gives $24 \equiv 0$), so a different idea is needed.
(Lebesgue's trick.) Add a constant that makes one side a sum of two squares and the other a sum of two cubes: $23 + 4 = 27$, so

$$y^2 + 2^2 = x^3 + 27 = (x + 3)(x^2 - 3x + 9).$$

Suppose $x \equiv 1 \pmod 4$. Then $x^2 - 3x + 9 \equiv 1 - 3 + 9 \equiv 3 \pmod 4$, and $x^2 - 3x + 9 = \left(x - \frac{3}{2}\right)^2 + \frac{27}{4} > 0$. A positive integer that is 3 modulo 4 has a prime factor $q \equiv 3 \pmod 4$, since a product of primes that are 1 modulo 4 is 1 modulo 4. Then $q \mid y^2 + 2^2$, and Lemma 8 gives $q \mid 2$, a contradiction. Hence there are no solutions.
The positivity check is needed: the step "a number that is 3 modulo 4 has a prime factor that is 3 modulo 4" is false for negative numbers such as -1.

Kevinvariant
kevinvariant08.github.io

1.5 Counting with the Legendre symbol

By Theorem 1, $\sum_{a=0}^{p-1}\left(\frac{a}{p}\right)=0$. Sums of the symbol over a quadratic are almost as simple.

Lemma 10.
If $p\nmid c$ then $\displaystyle\sum_{x=0}^{p-1}\left(\frac{x^2-c}{p}\right)=-1$.

Proof. Count the solutions of $x^2-y^2\equiv c$ in two ways. Writing $u=x-y$ and $v=x+y$, which is a bijection on pairs since 2 is invertible, we need $uv\equiv c$: u is any of the $p-1$ nonzero residues and then v is determined, so there are $p-1$ solutions. On the other hand, for each x there are $1+\left(\frac{x^2-c}{p}\right)$ values of y. Hence $p+\sum_x\left(\frac{x^2-c}{p}\right)=p-1$. □

Example 4.
Let p be an odd prime. How many n with $1\le n\le p-2$ are such that n and $n+1$ are both quadratic residues modulo p?

Solution.
We count with indicator functions: for $1\le a\le p-1$ the quantity $\frac{1}{2}\left(1+\left(\frac{a}{p}\right)\right)$ is 1 for a residue and 0 otherwise. So the required number is

$$N=\frac{1}{4}\sum_{n=1}^{p-2}\left(1+\left(\frac{n}{p}\right)\right)\left(1+\left(\frac{n+1}{p}\right)\right).$$

Expand into four sums. The first is $p-2$. The second is $\sum_{n=1}^{p-2}\left(\frac{n}{p}\right)=0-\left(\frac{p-1}{p}\right)=-\left(\frac{-1}{p}\right)$.
The third is $\sum_{m=2}^{p-1}\left(\frac{m}{p}\right)=-\left(\frac{1}{p}\right)=-1$.
The fourth is $\sum_{n=1}^{p-2}\left(\frac{n(n+1)}{p}\right)$. The terms $n=0$ and $n=p-1$ would be 0, so we may sum over all residues n. Since $4n(n+1)=(2n+1)^2-1$ and $\left(\frac{4}{p}\right)=1$, each term equals $\left(\frac{(2n+1)^2-1}{p}\right)$, and $2n+1$ runs over all residues, so Lemma 10 gives -1.
Therefore $N=\frac{1}{4}\left(p-4-\left(\frac{-1}{p}\right)\right)$, that is, $\frac{p-5}{4}$ if $p\equiv 1\pmod 4$ and $\frac{p-3}{4}$ if $p\equiv 3\pmod 4$.
Check: modulo 13 the residues are $1,3,4,9,10,12$ and the pairs are $(3,4)$ and $(9,10)$. In particular two consecutive nonzero residues exist for every $p\ge 7$.

Common traps.

- Euler's criterion and Theorems 3 to 5 are for odd primes. Treat $p=2$, and primes dividing the coefficients, separately; Example 2 shows how.
- Reciprocity flips $\left(\frac{q}{p}\right)$ only for odd primes $q\neq p$. Factor the top first: $\left(\frac{6}{p}\right)=\left(\frac{2}{p}\right)\left(\frac{3}{p}\right)$, and 2 uses Theorem 4.
- For composite moduli, being a square modulo n means being a square modulo every prime power dividing n. A product of symbols equal to 1 proves nothing: $\left(\frac{2}{3}\right)\left(\frac{2}{5}\right)=1$, but 2 is not a square modulo 15.
- Lemma 8 says q divides both x and y, not that q cannot divide x^2+y^2: $3\mid 3^2+6^2$.
- "A number $\equiv 3\pmod 4$ has a prime factor $\equiv 3\pmod 4$" needs the number to be positive.

Kevinvariant
kevinvariant08.github.io

1.6 How to spot this in a contest

The signs that quadratic residues are relevant, in rough order of frequency:

1. A prime p divides an expression that can be rearranged as "$a \equiv$ square", for instance $p \mid n^2 + k$, $p \mid 4n^2 - 3$, or $p \mid 2^{\text{odd}} + 1$ as in the opening. Translate to $\left(\frac{a}{p}\right) = 1$ and compute with reciprocity; the conclusion is a congruence condition on p.
2. An exponent $\frac{p-1}{2}$ appears anywhere: that is Euler's criterion.
3. An equation such as $y^2 = x^3 + k$: look for a constant that turns it into "sum of two squares $=$ product", then find a factor that is 3 modulo 4, or more generally a prime for which some number is a non-residue.
4. Sums of two squares, or $x^2 + y^2$ divisible by something: Lemma 8 and Theorem 9.
5. Counting residues with a condition attached: indicator functions and Lemma 10.
6. "Infinitely many primes p such that $\left(\frac{a}{p}\right) = 1$": a Euclid-type argument with a quadratic polynomial, see L5-NT2.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ Find all primes $p > 3$ for which the congruence $x^2 \equiv 6 \pmod{p}$ has a solution. Give the answer as a list of residues modulo 24.

Problem 2 ⋆⋆ Let a be an integer and b an odd integer. Prove that $b^2 + 2$ does not divide $a^2 + 1$. Show by an example that the condition that b is odd cannot be dropped.

Problem 3 ⋆⋆⋆ The number 2027 is prime. Find the number of ordered pairs (x, y) of integers with $0 \le x, y \le 2026$ such that 2027 divides $x^2 + 5y^2 - 1$.

Problem 4 ⋆⋆⋆ Let p be a prime with $p \equiv 3 \pmod{8}$. Prove that $p = x^2 + 2y^2$ for some odd positive integers x and y.

Problem 5 ⋆⋆⋆ Let n be a positive integer and let $N = \frac{3^{2^n}+1}{2}$.

(a) Prove that every prime factor p of N satisfies $p \equiv 1 \pmod{2^{n+1}}$, and that $p \equiv 1 \pmod{2^{n+2}}$ if and only if $p \equiv 1 \pmod{3}$.

(b) Prove that N has a prime factor p with $p \equiv 2^{n+1} + 1 \pmod{2^{n+2}}$.

Problem 6 ⋆⋆⋆⋆ Let p be a prime with $p \equiv 3 \pmod{8}$, and write $p = x^2 + 2y^2$ with x and y positive integers (practice problem 4 shows that this is possible). Prove that

(a) x is a quadratic residue modulo p if and only if $x \equiv 1$ or $3 \pmod{8}$;

(b) y is a quadratic residue modulo p if and only if $y \equiv 1 \pmod{4}$.

Kevinvariant
kevinvariant08.github.io

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT BMO1 2020/21 Problem 6** ⋆⋆
   Pin down the possible Mersenne primes from the first condition, then use the criterion for sums of two squares.

2. **IMO 1996 Problem 4** ⋆⋆
   Combine the two squares to get a sum of two fourth powers divisible by 481 = 13 x 37, then ask which primes can divide such a sum.

3. **EGMO 2016 Problem 6** ⋆⋆⋆
   Reduce to a few Pell-type equations, then compare with the squares modulo 7.

4. **UKMT BMO2 2015/16 Problem 4** ⋆⋆⋆
   The condition says $u^2 + v^2 = 2p^2$; rewrite it as a sum of two squares equal to a square and use what you know about representations of p.

5. **UKMT BMO2 2013/14 Problem 3** ⋆⋆⋆
   For part (b), ask what p dividing a term forces on the previous term, and complete the square.

6. **USAMO 2008 Problem 1** ⋆⋆⋆
   Aim for a product of the form $a^2 + a + 1$; which primes divide such numbers?

7. **IMO 2008 Problem 3** ⋆⋆⋆⋆
   Work backwards from a large prime p = 1 mod 4 and the square roots of -1 modulo p.

8. **USAMO 2020 Problem 3** ⋆⋆⋆⋆
   Parametrise the a for which a and 4 - a have prescribed residue behaviour, treating p mod 4 separately.

9. **USAMO 2026 Problem 6** ⋆⋆⋆⋆
   Use the parity of the totient to restrict the shape of ab + 1; reciprocity for -3 appears in one case.

Kevinvariant
kevinvariant08.github.io

4 Hints

1. Use $\left(\frac{6}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{3}{p}\right)$; the two factors depend on p modulo 8 and modulo 12.
2. Work out $b^2 + 2$ modulo 8 for odd b. Which primes can divide $a^2 + 1$?
3. For fixed y, the number of suitable x is $1 + \left(\frac{1-5y^2}{p}\right)$. Pull out a constant so that Lemma 10 applies.
4. -2 is a residue modulo p. Copy the pigeonhole proof of Theorem 7, then use parity and the residue of p modulo 8.
5. For (a), find $\operatorname{ord}_p(3)$, then evaluate $3^{(p-1)/2}$ with Euler's criterion and reciprocity. For (b), look at N modulo 3.
6. For an odd prime $q \mid x$, reduce $p = x^2 + 2y^2$ modulo q to find $\left(\frac{p}{q}\right)$, then flip with reciprocity, remembering that $p \equiv 3 \pmod 4$. Then multiply over the prime factors of x.

5 Solutions

Problem 1

Find all primes $p > 3$ for which the congruence $x^2 \equiv 6 \pmod p$ has a solution. Give the answer as a list of residues modulo 24.

Since $p > 3$ we have $p \nmid 6$, and we need $\left(\frac{6}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{3}{p}\right) = 1$, so the two symbols must be equal. By Theorem 4, $\left(\frac{2}{p}\right) = 1$ exactly when $p \equiv \pm 1 \pmod 8$, and by the formula from the lesson, $\left(\frac{3}{p}\right) = 1$ exactly when $p \equiv \pm 1 \pmod{12}$. Both conditions depend only on p modulo 24, and p is one of $1, 5, 7, 11, 13, 17, 19, 23$ modulo 24.

Going through these: 1 gives $(+,+)$; 5 gives $(-,-)$; 7 gives $(+,-)$; 11 gives $(-,+)$; 13 gives $(-,+)$; 17 gives $(+,-)$; 19 gives $(-,-)$; 23 gives $(+,+)$. The symbols agree for $p \equiv 1, 5, 19, 23 \pmod{24}$, and these are the required primes.

Answer: $p \equiv 1, 5, 19, 23 \pmod{24}$, that is, $p \equiv \pm 1$ or $\pm 5 \pmod{24}$.

Problem 2

Let a be an integer and b an odd integer. Prove that $b^2 + 2$ does not divide $a^2 + 1$. Show by an example that the condition that b is odd cannot be dropped.

Since b is odd, $b^2 \equiv 1 \pmod 8$, so $b^2 + 2 \equiv 3 \pmod 8$. Thus $b^2 + 2$ is odd, at least 3 and congruent to 3 modulo 4. A product of odd primes that are all 1 modulo 4 is 1 modulo 4, so $b^2 + 2$ has a prime factor $q \equiv 3 \pmod 4$.

Suppose $b^2 + 2 \mid a^2 + 1$. Then $q \mid a^2 + 1$, so $a^2 \equiv -1 \pmod q$, and $q \nmid a$ since $q \nmid 1$. So -1 is a quadratic residue modulo q, which contradicts Theorem 3 as $q \equiv 3 \pmod 4$. Hence $b^2 + 2 \nmid a^2 + 1$.

For even b the statement can fail. Take $b = 12$: then $b^2 + 2 = 146 = 2 \cdot 73$, and $73 \equiv 1 \pmod 4$. Indeed $27^2 + 1 = 730 = 146 \cdot 5$, so $a = 27$ works. (For $b = 2, 4, 6, 8, 10$ the number $b^2 + 2$ is

Kevinvariant
kevinvariant08.github.io

$6, 18, 38, 66, 102$, and each has a prime factor 3 or 19, which is 3 modulo 4, so 12 is the smallest positive even example.) □

Problem 3

The number 2027 *is prime. Find the number of ordered pairs* (x, y) *of integers with* $0 \le x, y \le 2026$ *such that* 2027 *divides* $x^2 + 5y^2 - 1$.

Write $p = 2027$. For fixed y the number of x with $x^2 \equiv 1 - 5y^2$ is $1 + \left(\frac{1-5y^2}{p}\right)$, so the number of pairs is

$$N = p + \sum_{y=0}^{p-1} \left(\frac{1-5y^2}{p}\right).$$

Let c be the inverse of 5 modulo p. Then $1 - 5y^2 \equiv -5(y^2 - c)$, and by multiplicativity each term is $\left(\frac{-5}{p}\right)\left(\frac{y^2-c}{p}\right)$. Since $p \nmid c$, Lemma 10 gives $\sum_y \left(\frac{y^2-c}{p}\right) = -1$, so $N = p - \left(\frac{-5}{p}\right)$.

Now $p = 2027 \equiv 3 \pmod 4$, so $\left(\frac{-1}{p}\right) = -1$, and by reciprocity $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right) = \left(\frac{2}{5}\right) = -1$, as $2027 \equiv 2 \pmod 5$ and the squares modulo 5 are 1 and 4. Hence $\left(\frac{-5}{p}\right) = (-1)(-1) = 1$ and $N = 2027 - 1 = 2026$.

Answer: 2026

Problem 4

Let p *be a prime with* $p \equiv 3 \pmod 8$. *Prove that* $p = x^2 + 2y^2$ *for some odd positive integers* x *and* y.

By Theorem 4, $\left(\frac{-2}{p}\right) = 1$, so there is r with $r^2 \equiv -2 \pmod p$. Let $m = \lfloor \sqrt{p} \rfloor$, so $m < \sqrt{p}$ as p is not a square. Among the $(m+1)^2 > p$ pairs (a, b) with $0 \le a, b \le m$, two different pairs give the same residue of $a + br$, say $a + br \equiv a' + b'r$. Put $u = a - a'$ and $v = b' - b$. Then $u \equiv rv$, so $u^2 \equiv -2v^2$ and $p \mid u^2 + 2v^2$. Also $(u, v) \neq (0, 0)$ and $|u|, |v| < \sqrt{p}$, so $0 < u^2 + 2v^2 < 3p$. Hence $u^2 + 2v^2$ equals p or $2p$.

If $u^2 + 2v^2 = 2p$ then u is even, $u = 2w$, and dividing by 2 gives $v^2 + 2w^2 = p$. So in both cases $p = x^2 + 2y^2$ for some integers x, y, which we may take to be non-negative. Since p is odd, x is odd. Also $y \neq 0$, since p is not a square. If y were even then $2y^2 \equiv 0 \pmod 8$ and $p \equiv x^2 \equiv 1 \pmod 8$, contradicting $p \equiv 3 \pmod 8$. Hence x and y are both odd and positive. □

Remark. For example $3 = 1 + 2$, $11 = 9 + 2$, $19 = 1 + 18$, $43 = 25 + 18$.

Problem 5

Let n *be a positive integer and let* $N = \frac{3^{2^n}+1}{2}$.

(a) Prove that every prime factor p *of* N *satisfies* $p \equiv 1 \pmod{2^{n+1}}$, *and that* $p \equiv 1 \pmod{2^{n+2}}$ *if and only if* $p \equiv 1 \pmod 3$.

(b) Prove that N *has a prime factor* p *with* $p \equiv 2^{n+1} + 1 \pmod{2^{n+2}}$.

(a) Since 3^{2^n} is odd, N is an integer and $2N = 3^{2^n} + 1$. As $n \ge 1$, $3^{2^n} = 9^{2^{n-1}} \equiv 1 \pmod 8$, so $2N \equiv 2 \pmod 8$ and N is odd. Also $2N \equiv 1 \pmod 3$, so $3 \nmid N$. Let p be a prime factor of N; then $p \neq 2, 3$.

From $3^{2^n} \equiv -1 \pmod p$ we get $3^{2^{n+1}} \equiv 1$ and $3^{2^n} \not\equiv 1$, as $p \neq 2$. So $\mathrm{ord}_p(3)$ divides 2^{n+1} but not 2^n, that is, $\mathrm{ord}_p(3) = 2^{n+1}$. By Fermat, $2^{n+1} \mid p - 1$. In particular $p \equiv 1 \pmod 4$.

Kevinvariant
kevinvariant08.github.io

By Euler's criterion $3^{(p-1)/2} \equiv \left(\frac{3}{p}\right) \pmod p$, and since $p \equiv 1 \pmod 4$, reciprocity gives $\left(\frac{3}{p}\right) = \left(\frac{p}{3}\right)$. If $p \equiv 1 \pmod 3$, then $3^{(p-1)/2} \equiv 1$, so the order 2^{n+1} divides $\frac{p-1}{2}$, that is, $2^{n+2} \mid p-1$. If $p \equiv 2 \pmod 3$, then $3^{(p-1)/2} \equiv -1 \not\equiv 1$, so $2^{n+1} \nmid \frac{p-1}{2}$ and $2^{n+2} \nmid p-1$. This proves both directions.

(b) Since $2N \equiv 1 \pmod 3$, we have $N \equiv 2 \pmod 3$. Every prime factor of N is 1 or 2 modulo 3, and if all were 1 modulo 3 then N would be too. So N has a prime factor $p \equiv 2 \pmod 3$. By (a), $2^{n+1} \mid p-1$ and $2^{n+2} \nmid p-1$, so $\frac{p-1}{2^{n+1}}$ is odd, which means $p \equiv 2^{n+1}+1 \pmod{2^{n+2}}$. □

Remark. Part (b) can also be proved by LTE (L4-NT1): $v_2\left(3^{2^n}-1\right) = n+2$, so $N-1 = \frac{3^{2^n}-1}{2}$ is 2^{n+1} times an odd number, while a product of numbers that are 1 modulo 2^{n+2} is 1 modulo 2^{n+2}. Part (a) needs reciprocity. For $n=3$, $N = 3281 = 17 \cdot 193$, with $17 = 2^4+1$ and $193 = 3 \cdot 2^6+1$.

Problem 6

Let p be a prime with $p \equiv 3 \pmod 8$, and write $p = x^2+2y^2$ with x and y positive integers (practice problem 4 shows that this is possible). Prove that

(a) x is a quadratic residue modulo p if and only if $x \equiv 1$ or $3 \pmod 8$;

(b) y is a quadratic residue modulo p if and only if $y \equiv 1 \pmod 4$.

Since p is odd, x is odd. If y were even then $8 \mid 2y^2$ and $p \equiv x^2 \equiv 1 \pmod 8$, which is false; so y is odd. A common prime factor r of x and y would give $r^2 \mid p$, so $\gcd(x,y) = 1$. Also $x, y < \sqrt{p} < p$, so $p \nmid x$ and $p \nmid y$. Since $p \equiv 3 \pmod 4$, reciprocity for an odd prime $q \neq p$ reads $\left(\frac{q}{p}\right) = (-1)^{(q-1)/2}\left(\frac{p}{q}\right) = \left(\frac{-1}{q}\right)\left(\frac{p}{q}\right)$, using Theorem 3.

Step 1. Let q be a prime factor of x. Then q is odd, $q \neq p$ and $q \nmid y$. Reducing modulo q, $p \equiv 2y^2$, so $\left(\frac{p}{q}\right) = \left(\frac{2}{q}\right)\left(\frac{y}{q}\right)^2 = \left(\frac{2}{q}\right)$. Hence $\left(\frac{q}{p}\right) = \left(\frac{-1}{q}\right)\left(\frac{2}{q}\right) = \left(\frac{-2}{q}\right)$, which by Theorem 4 is 1 if $q \equiv 1,3 \pmod 8$ and -1 if $q \equiv 5,7 \pmod 8$.

Step 2. Let q be a prime factor of y. Then q is odd, $q \neq p$ and $q \nmid x$. Reducing modulo q, $p \equiv x^2$, so $\left(\frac{p}{q}\right) = 1$ and $\left(\frac{q}{p}\right) = \left(\frac{-1}{q}\right)$, which is 1 if $q \equiv 1 \pmod 4$ and -1 if $q \equiv 3 \pmod 4$.

Step 3. For odd m let $\alpha(m) = 1$ if $m \equiv 1,3 \pmod 8$ and $\alpha(m) = -1$ if $m \equiv 5,7 \pmod 8$, and let $\beta(m) = (-1)^{(m-1)/2}$. Both are multiplicative on odd integers. For α this is a finite check of the products of $1,3,5,7$ modulo 8: $3 \cdot 3 \equiv 1$, $3 \cdot 5 \equiv 7$, $3 \cdot 7 \equiv 5$, $5 \cdot 5 \equiv 1$, $5 \cdot 7 \equiv 3$, $7 \cdot 7 \equiv 1$. For β, write $mm' - 1 = (m-1)(m'-1) + (m-1) + (m'-1)$; the first term is divisible by 4, so $\frac{mm'-1}{2} \equiv \frac{m-1}{2} + \frac{m'-1}{2} \pmod 2$.

Write $x = q_1 q_2 \cdots q_s$ as a product of (not necessarily distinct) primes, with $s = 0$ if $x = 1$. By multiplicativity of the Legendre symbol and Step 1,

$$\left(\frac{x}{p}\right) = \prod_{i=1}^{s}\left(\frac{q_i}{p}\right) = \prod_{i=1}^{s}\alpha(q_i) = \alpha(x).$$

Since $p \nmid x$, x is a residue if and only if $\alpha(x) = 1$, that is, if and only if $x \equiv 1$ or $3 \pmod 8$. In the same way Step 2 gives $\left(\frac{y}{p}\right) = \beta(y)$, so y is a residue if and only if $y \equiv 1 \pmod 4$. □

Check. $11 = 3^2 + 2 \cdot 1^2$ and $3 \equiv 5^2 \pmod{11}$; $43 = 5^2 + 2 \cdot 3^2$, and both 5 and 3 are non-residues modulo 43; $59 = 3^2 + 2 \cdot 5^2$, and $3 \equiv 11^2$, $5 \equiv 8^2 \pmod{59}$.

Kevinvariant
kevinvariant08.github.io